MARK SCHEME · TEACHER COPY

12BR · Topic 15: Hardware

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Total marks: 62

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 62 marks

Question 1 · Oct/Nov 2025 Paper 31, original Q5 (a),(b) [8 marks]

Question	Answer	Marks
1(a)	**Two** marks for all **six** correct terms **One** mark for any **three** correct terms $Z = \bar{A}.\bar{B}.\bar{C}.\bar{D} + \bar{A}.B.\bar{C}.D + \bar{A}.B.C.D + A.\bar{B}.\bar{C}.\bar{D} + A.B.\bar{C}.D + A.B.C.D$	2
1(b)(i)	**Two** marks if no errors present **One** mark if one error present AB / CD 00: 1 0 0 1 01: 0 1 1 0 11: 0 1 1 0 10: 0 0 0 0	2

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Total: 62 marks

1(b)(ii)	**One** mark for each correct loop AB / CD 00 01 11 10 00: 1 0 0 1 01: 0 1 1 0 11: 0 1 1 0 10: 0 0 0 0	**2**
1(b)(iii)	**One** mark for each mark point • One correct Boolean term • Boolean terms and operator correct and no other terms present **Z =** B.D + $\overline{B}.\overline{C}.\overline{D}$ // $\overline{B}.\overline{C}.\overline{D}$ + B.D	**2**

Question 2 · May/June 2026 Paper 33, original Q4 (a),(b) [5 marks]

Question	Answer	Marks	Guidance
2 ·	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.		
2(a)	**One** mark per mark point (**Max 3**) MP1 Massively parallel computers involve the **linking together** of **many processors / computers** MP2 … to form a more powerful 'single machine' MP3 All the processors work simultaneously on the same problem MP4 … with each processor being responsible for its **own part of the problem** MP5 … and communication being via interconnected data pathways/networks // … and communicate via shared pools of memory.	**3**	Accept many/multiple/thousands/extremely large number for 'many' but not some/several.

Total: 62 marks

2(b)	**One** mark for a difference (**Max 1**) MP1 SIMD systems all execute the same instruction and MISD systems perform different instructions MP2 SIMD systems work on many data streams/elements simultaneously MISD systems all work on the same data source/element MP3 MISD systems are more complex than SISD systems // allow the reverse **One** mark for a similarity (**Max 1**) MP4 SIMD and MISD are both parallel systems MP5 SIMD and MISD using multiple processors MP6 SIMD and MISD both may be pipelined MP7 Neither SIMD nor MISD are well suited for sequential applications such as office productivity software or compilers	**2**	Both SIMD and MISD must appear or be clearly inferred in each response for the mark. Simply expanding the acronym is NE.

2	W	N3	7	13	20

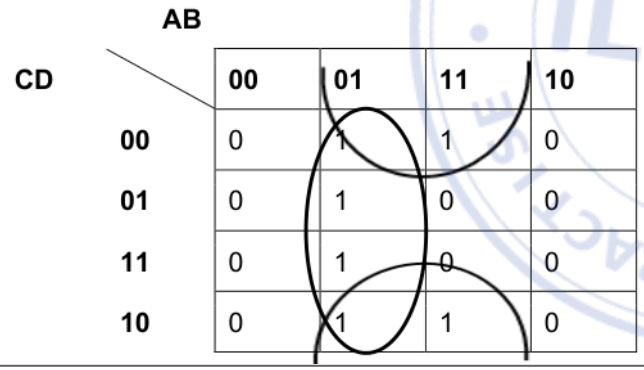
Question 3 · May/June 2026 Paper 33, original Q6 (a),(b) [8 marks]

Question	Answer	Marks	Guidance
3	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Annotation	Meaning
✓	Correct
✗	Incorrect
λ	To indicate where a key word/phrase/code is missing.
≈	Not relevant or used to separate parts of an answer.
~~	Indicates a part of the answer that is incorrect.
Highlighter	To draw attention to a particular aspect or to indicate where parts of an answer have been combined.

TV	Too vague.
REP	Repetition
NE	No examples or not enough.

Annotation	**Meaning**
BOD	Benefit of Doubt.
NAQ	Not Answered Question.
SEEN	Indicates that work or a page has been seen including blank answer spaces and blank pages.
FT	Follow through.
I	Ignore

3(a)(i)	**One** mark for each correct loop **(Max 2)** 	**2**	If incorrect loops seen, max 1.
3(a)(ii)	**One** mark per mark point (**Max 2**) MP1 One correct Boolean term MP2 Both correct Boolean terms connected by an OR sign and no other terms present $\bar{A}.B + B.\bar{D}$	**2**	Terms can be in either order Accept answers as AND if no dot e.g. A.B = AB $\bar{A} \cdot B + B \cdot \bar{C} \cdot \bar{D} + B \cdot C \cdot \bar{D}$

3(a)(iii)	B.(A + D) // B.(A.D)	1	Terms can be in any order Allow answers with or without dot e.g. AB for A.B
3(b)	**One** mark for correct answer **(Max 2)** for correct working from points shown **X = ~~A + B + C + B + C~~** **(X =) $\overline{A}.\overline{B}.\overline{C} + \overline{B}.\overline{C}$** — De Morgan's laws **(X =) $\overline{B}.\overline{C}.(\overline{A} + 1)$** — Distributive law Correct answer — Annulment law **(X =) $\overline{B}.\overline{C}$ // $\overline{B + C}$**	3	One mark for correct answer and up to two marks to show correct use of different laws in the solution. The top line, shaded grey, is given in the question.

3	W⇨N2⇨N5	N4	4+7+2=13	9	22

Question 4 · May/June 2026 Paper 32, original Q5 [4 marks]

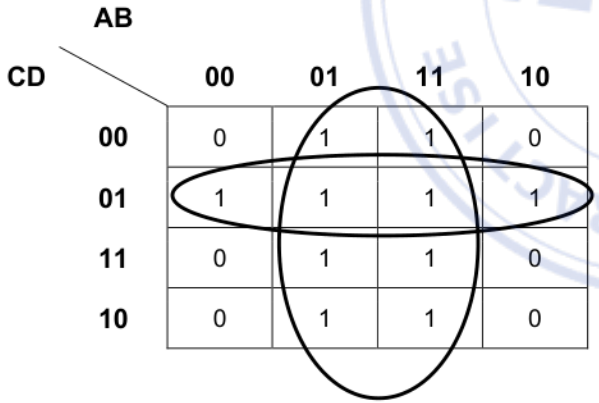
Question	Answer	Marks	Guidance
4	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.		
4	**One** mark per mark point (**Max 4**) MP1 A VM is a standalone environment running on a physical computer // A VM is not a physical machine MP2 It can emulate the hardware/software **of another system** MP3 VM software has all the components of a physical computer system MP4 The **host** machine provides all the physical resources e.g. RAM/ROM MP5 … and is controlled via a **host** operating system MP6 The hypervisor runs between the **host** and **guest** operating systems MP7 … and creates and runs virtual machines / allocates resources from the **host** to the **guest** OS. MP8 A **guest** OS runs within the VM/**host** OS MP9 Multiple **guest** OS can run on the same VM // Multiple VMs can run on the same **host** OS MP10 The **guest** operating system controls/accesses the **virtual** hardware/software.	4	Do not allow benefits, drawbacks or uses, as the question is about the meaning of the term virtual machine, so requires a description.

Total: 62 marks

Question 5 · May/June 2026 Paper 32, original Q7 (a),(b) [8 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>5(a)(i)</td><td>Two marks if no errors present
One mark if one error present

<table>
<tr><th>CD \ AB</th><th>00</th><th>01</th><th>11</th><th>10</th></tr>
<tr><td>00</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>01</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>11</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>10</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
</table></td><td>2</td><td>0’s must be present</td></tr>
<tr><td>5(a)(ii)</td><td>One mark for each correct loop (Max 2)


</td><td>2</td><td>Max 1 if incorrect loops present.</td></tr>
<tr><td>5(a)(iii)</td><td>B + $\overline{C}$.D // $\overline{C}$.D + B</td><td>1</td><td>Allow products with or without dots
e.g. C.D or CD.</td></tr>
</table>

5(b)	**One** mark for correct answer **(Max 2)** for correct working from points shown $X = \overline{\overline{A}.B.C.\overline{D}} + \overline{\overline{A}.D}$ De Morgan's laws $(X =) \overline{\overline{A}} + \overline{B} + \overline{C} + \overline{\overline{D}} + \overline{\overline{A}} . \overline{\overline{D}}$ // $(X =) \overline{\overline{A}} + \overline{B} + \overline{C} + \overline{\overline{D}} + A.D$ Double negation law $(X =) A + \overline{B} + \overline{C} + D + A.D$ Absorption law Correct answer $(X =) A + \overline{B} + \overline{C} + D$	3	One mark for correct answer and up to two marks to show correct use of different laws in the solution, which could be in the same line. The top line, shaded grey, is given in the question. Other solutions are possible. If distributive law is seen, that is also worth 1 working mark. Mark correct working lines where you see them.

Question 6 · Oct/Nov 2025 Paper 32, original Q7 (a),(b),(c) [8 marks]

Question	Answer	Marks
6(a)	**One** mark for working, (all four columns P, Q, R and S) **One** mark for first eight rows of column Z (Shaded) **One** mark for second eight rows of column Z (Unshaded)	3

				Working space				
A	B	C	D	P	Q	R	S	Z
0	0	0	0	1	0	0	1	0
0	0	0	1	1	0	0	1	0
0	0	1	0	0	1	0	1	0
0	0	1	1	0	1	0	1	0
0	1	0	0	1	0	0	1	0
0	1	0	1	1	0	0	1	0
0	1	1	0	0	0	0	0	1
0	1	1	1	0	0	1	0	0
1	0	0	0	1	0	0	1	0
1	0	0	1	1	0	0	1	0
1	0	1	0	0	0	0	0	1
1	0	1	1	0	0	0	0	1
1	1	0	0	1	0	0	1	0
1	1	0	1	1	0	0	1	0
1	1	1	0	0	0	0	0	1
1	1	1	1	0	0	1	0	0

6(b)	**One** mark for correct answer (terms may be in any order) $Z = \bar{A}.B.C.\bar{D} + A.\bar{B}.C.\bar{D} + A.\bar{B}.C.D + A.B.C.\bar{D}$	**1**
6(c)	**One** mark for each mark point • Correct application of De Morgan's laws • Correct application of Idempotent, Distributive or Absorption laws • Correct application of Idempotent, Distributive or Absorption laws • Correct final answer $\overline{(A+B).(A.B+B.C)}$ $(\bar{A}.\bar{B}).(\bar{A}+\bar{B}+\bar{B}+\bar{C})$ DeMorgan's laws $(\bar{A}.\bar{B}).(\bar{A}+\bar{B}+\bar{C})$ Idempotent laws $(\bar{A}.\bar{A}+\bar{A}.\bar{B}+\bar{A}.\bar{C}).(\bar{B}.\bar{A}+\bar{B}.\bar{B}+\bar{B}.\bar{C})$ Distributive laws $(\bar{A}+\bar{A}.\bar{B}+\bar{A}.\bar{C}).(\bar{A}.\bar{B}+\bar{B}+\bar{B}.\bar{C})$ Idempotent laws $(\bar{A}+\bar{A}.\bar{C}).(\bar{B}+\bar{B}.\bar{C})$ Absorption laws $\bar{A}.\bar{B}$ // $\overline{A + B}$ Absorption laws / Final Answer	**4**

Question 7 · May/June 2026 Paper 31, original Q5 [4 marks]

Question	Answer	Marks	Guidance
7	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.		

7	**One** mark for each correct benefit (**Max 2**) MP1 Virtual machines can provide an instruction set architecture that is different to the actual computer. // Multiple operating system environments can be used on the same computer. // You can run an application / legacy software which is not compatible with the host computer / operating system using a guest operating system that is compatible with the application / software. MP2 Costs can be saved by not having to purchase additional hardware to test / use applications. MP3 The guest operating system can crash without affecting the host machine. // Guest operating system hosted on a virtual machine can be used without impacting anything outside the virtual machine. // When trying a piece of suspicious software, if it causes a problem, it will only infect the virtual machine. **One** mark for each correct limitation (**Max 2**) MP4 Degraded performance. // A virtual machine is less efficient than real machines because they access hardware indirectly. // The virtual machine has to request access to the hardware from the host, which slows usability. // If multiple virtual machines are running on the same host, performance may be negatively affected if the host lacks sufficient power to deal with all their requirements. MP5 A virtual machine is dependent on the features / standard of the host machine. // It may be affected by the weaknesses of the host machine // Not all hardware/software can be emulated. MP6 Managing and maintaining a virtual machine can be quite complex /specialised.	**4**	MP2 do not allow the mark for 'virtual machines can be cheaper', 'cost saving' by itself, or similar. The answer must match the phrase in the mark scheme. MP6 do not allow for just saying 'virtual machine can be expensive'. It has to have a reason e.g. expensive to maintain.

Question 8 · May/June 2026 Paper 31, original Q6 (a),(b),(c) [9 marks]

Question	Answer	Marks	Guidance
8	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Annotation	Meaning
✓	Correct
✗	Incorrect

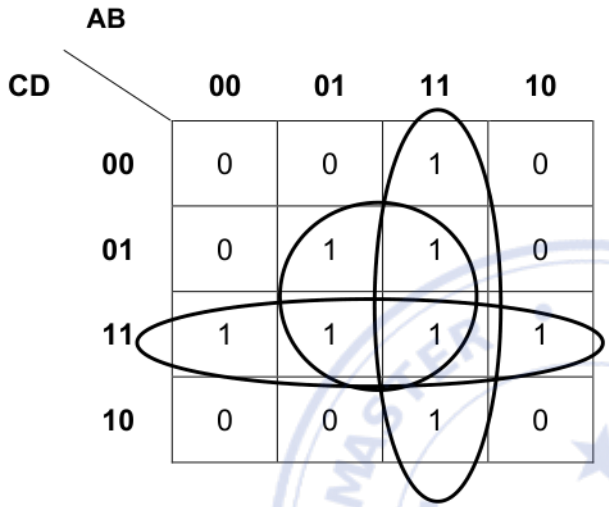
λ	To indicate where a key word/phrase/code is missing.
	Not relevant or used to separate parts of an answer.
	Indicates a part of the answer that is incorrect.
Highlighter	To draw attention to a particular aspect or to indicate where parts of an answer have been combined.
TV	Too vague.
REP	Repetition
NE	No examples or not enough.
Annotation	**Meaning**
BOD	Benefit of Doubt.
NAQ	Not Answered Question.
SEEN	Indicates that work or a page has been seen including blank answer spaces and blank pages.
FT	Follow through.
I	Ignore

8(a)	• **One** mark for columns P, Q and R • **One** mark for column S • **One** mark for column Z	**3**	

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				Working space						
	A	**B**	**C**	**P**	**Q**	**R**	**S**	**Z**		
	0	0	0	1	1	1	0	0		
	0	0	1	1	1	0	1	0		
	0	1	0	1	0	1	0	0		
	0	1	1	1	0	1	1	0		
	1	0	0	0	1	1	1	1		
	1	0	1	0	1	1	0	0		
	1	1	0	0	0	1	1	0		
	1	1	1	0	0	1	0	0		

8(b)	**One** mark per mark point (**Max 1**) Solving logic expressions using a K-map makes it easier because you don't need to use Boolean algebra. // It is **guaranteed** to give you an optimal solution. // Converts complex logic problems into simpler ones.	**1**	Makes it easier by itself is not enough. Do not allow optimal sum-of-products instead of optimal solution.

8(c)(i)	**One** mark for each correct loop **(Max 3)**  	**3**	Max 2 if incorrect loops seen.
8(c)(ii)	**One** mark per mark point (**Max 2**) MP1 One correct Boolean term with an OR (+) sign MP2 All three correct Boolean terms connected by OR signs and no other terms present. A.B + B.D + C.D	**2**	Terms can be in any order.

Question 9 · Oct/Nov 2025 Paper 33, original Q3 (a),(b) [8 marks]

Question	Answer	Marks
9(a)	**Two** marks for all **six** correct terms **One** mark for any **four** correct terms $Z = \bar{A}.\bar{B}.C.\bar{D} + \bar{A}.\bar{B}.C.D + \bar{A}.B.\bar{C}.\bar{D} + A.\bar{B}.C.\bar{D} + A.\bar{B}.C.D + A.B.\bar{C}.\bar{D}$	**2**

9(b)(i)	**Two** marks if no errors present **One** mark if one error present **AB** **CD** | **00** | **01** | **11** | **10** **00** | 0 | 1 | 1 | 0 **01** | 0 | 0 | 0 | 0 **11** | 1 | 0 | 0 | 1 **10** | 1 | 0 | 0 | 1	**2**
9(b)(ii)	**One** mark for each correct loop **(Max 2)** **AB** **CD** | **00** | **01** | **11** | **10** **00** | 0 | 1 | 1 | 0 **01** | 0 | 0 | 0 | 0 **11** | 1 | 0 | 0 | 1 **10** | 1 | 0 | 0 | 1	**2**
9(b)(iii)	**One** mark for each mark point • One correct Boolean term • Boolean terms and operator correct and no other terms present $Z = \overline{B}.C + B.\overline{C}.\overline{D}$ // $B.\overline{C}.\overline{D} + \overline{B}.C$	**2**

	9											

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